A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 37

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Yield increases as inputs increase **1** But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Second unit:** 70 - 50 = 20 units **1 Fifth unit:** 94 - 90 = 4 units **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (units) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** The first units of input on each area give the largest yield increases **1** Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Some areas have much greater potential productivity so investment there is sensible **1** Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Yield per unit area is very high because of large inputs **1** But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Sieves assembled with the coarsest mesh on top – a dried, crushed soil sample is placed in the top sieve **1** Stack shaken (about 2 minutes) until no more particles pass – each fraction weighed **1** Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated – material over 2 mm (stones, twigs) is not soil – plotted on a soil triangle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Auger collects a core of known cross-sectional area and depth **1** So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals such as calcium carbonate break down releasing carbon dioxide **1** So mass is lost that is not organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** A beam of light passes through the sample in a dark chamber – a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter **1** Standardise bulb brightness / colour, distance and container **1** A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations – Secchi discs / turbidity bottles give only semi-quantitative results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity **1** Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) **1** Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) **1** They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) **1** Many are in regions where environmental degradation / habitat loss threatens the wild plants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** A large population may all descend from a few closely related individuals so they are genetically similar **1** Similar adaptations mean all are susceptible to the same change (disease, climate) **1** Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs) – so species should be protected across their whole range to retain regional genes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
04.4	1 mark for the point below. **1** Morama bean (drought resistant, soya-like protein) – yeheb tree (drought resistant nuts, poor soils) – potato bean (high protein) – perennial maize / Kernza (no annual replanting, less ploughing / erosion) – salt-tolerant spinach relatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 2 400 / 2 000 **1** = 1.2 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Carnivorous species have a high market value, so they are the most profitable to farm, and consumers in wealthy countries prefer them **1** They are fed on fishmeal and fish oil made from wild-caught fish, so farming them still depends on catching wild fish and does not take the pressure off marine stocks **1** Several tonnes of wild fish may be needed per tonne of salmon, so the extra trophic level means the total amount of fish protein available falls rather than rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
05.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Uneaten feed and faeces fall through the cages and accumulate on the sea bed, where their decomposition removes oxygen and smothers the benthic community **1** The nutrients released cause local eutrophication and can trigger algal blooms **1** Parasites, especially sea lice, build up in the dense stock and spread to wild salmon and sea trout passing the cages **1** Antibiotics, pesticides used to treat sea lice and antifouling chemicals from the nets disperse into the water and affect non-target species **1** Escaped farmed fish interbreed with wild populations, diluting the locally adapted gene pool, and compete with them for food and spawning sites **1** Predators such as seals and cormorants attracted to the cages are shot or trapped **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Integrated multi-trophic aquaculture:** seaweeds and filter feeders such as mussels are grown alongside the cages and take up the dissolved nutrients and particulate waste as their own food **1** Siting cages where the current is strong, and leaving sites fallow between cycles, so wastes are dispersed and the sea bed recovers **1** Lowering stocking density and vaccinating the fish, so less medication is needed, and using cleaner fish such as wrasse to remove sea lice instead of pesticides **1** Stronger, better maintained nets and predator exclusion nets, so fewer fish escape and predators do not have to be killed **1** Replacing part of the fishmeal in the feed with plant or insect protein, reducing the demand on wild fish stocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
05.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Farming carnivorous species still depends on wild fish for feed, so expanding it increases rather than reduces the catch of wild fish **1** Suitable sheltered coastal sites are limited, and they compete with other uses and with sensitive habitats such as mangroves, which shrimp farming has already destroyed on a large scale **1** Some valuable species, such as tuna, cannot yet be bred reliably in captivity, so farms are stocked with wild-caught young fish and the wild population is still exploited **1** The local pollution, disease and escape problems mean that simply scaling the industry up multiplies its impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Organic advantages:** often waste products, locally available and cheap – add humus improving structure and water retention – increase soil biota – slow release reduces leaching **up to 2 Organic disadvantages:** nutrient composition cannot be controlled – released slowly so needs long-term planning – bulky with high water content so expensive to transport – cannot be applied to a growing crop – odour / pathogens **up to 2 Inorganic advantages:** composition controlled to match crop needs – nutrients released rapidly – concentrated and easy to apply **up to 2 Inorganic disadvantages:** high energy (fossil fuel) manufacture (Haber process) – no organic matter so humus and biota decline – some toxic to worms – high solubility so leached causing eutrophication – raw materials (phosphate rock) finite **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	6
06.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Crops grown in a circulating nutrient solution (inert medium) in greenhouses – nutrient concentrations monitored and adjusted so nutrients are never limiting – abiotic factors controlled **up to 2** Roots are small (all in contact with nutrients) so more growth goes into the harvested crop – no soil-borne pathogens or weeds – crop stays fresh with roots **up to 2 Disadvantages:** high inputs of nutrients and energy (heating, lighting, pumping) – high level of technical knowledge – only economic for high-value crops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
07.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Small clearings were farmed for a few years then abandoned as nutrients declined – the forest recolonised and soil fertility recovered before the area was cleared again **1** Population growth shortens the gap between clearances so the forest and soil do not recover **1** Clearings enlarge and degrade – less regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Rainforest soils are thin and nutrient-poor (nutrients held in the biomass) – once cleared, fertility falls in a few years and yields drop, so more land is cleared **1** Abandoned land may not be recolonised if no forest survives nearby – erosion of exposed soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing **1** Reservoirs (eg Tucurui, 2850 km^2) flood large areas, block migratory fish and river dolphins – decomposing vegetation produces acidic, deoxygenated water – fluctuating downstream flows **1** Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees) – cloud forests lose humidity as clouds form higher so frogs and epiphytes decline – some models predict replacement by savannah **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt **1** In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15 - 6) / 6 x 100 **1** = 150% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cells shed by the newts into the water contain their DNA **1** Water samples are analysed for DNA sequences unique to the species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year **1** Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups – limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance – marking (rings, wing tags) • **DNA:** eDNA to detect presence (great crested newts, invasive carp) – DNA databases to identify regional populations and trace timber, ivory and fish • **Auditory monitoring:** bat detectors and sonograms for presence, abundance and activity • **Position monitoring:** radio, GPS and acoustic transmitters – geolocator tags for small birds – data recorders – carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • **Satellites:** visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil – GPS for tracking • **Evaluation:** greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging – indirect evidence and traditional methods still needed • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	9
	Do not credit • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	New technologies have transformed what can be monitored and how often. Image databases of unique markings - tiger stripes, whale-shark spots, dolphin fin scars - let individuals be recognised, giving territory size, movements, lifespan and social structure without repeated capture, though only for species with unique features. Motion-sensitive and infrared cameras and CCTV record activity day and night with far less disturbance than observers, especially at nests.	
	Genetics adds detection where nothing can be seen: eDNA in water reveals great crested newts or invasive carp from shed cells, and DNA databases identify which regional population timber, ivory or fish came from, supporting enforcement. Bat detectors and sonograms give presence, abundance and activity of species heard but not seen. Position monitoring uses radio, GPS and acoustic transmitters, or tiny geolocator tags for birds too small to carry a transmitter, while data recorders on drones, AUVs, balloons, aircraft, satellites or the animals themselves collect abiotic data. Satellites map land use and deforestation in visible light, vegetation density and temperature in infrared, and wind, waves, sea level and oil slicks with radar that works through cloud.	
	The gains are coverage, continuity and reduced disturbance. The limits are cost, battery life and weight, the need to calibrate sensors and ground-truth remote data, ethical questions about tagging, and the fact that many species still require traditional methods and indirect evidence. New technology therefore complements rather than replaces fieldwork, but for rare, remote or nocturnal species it has made effective conservation possible where it was not before.	
	Weaker answer - Level 1 (3 marks)	
	New technologies help monitor wildlife. Cameras can take pictures of animals at night. GPS collars can track where animals go. Satellites can see deforestation from space. DNA can tell which animals are present. Bat detectors can hear bats. These help conservation because we know where animals are and how many there are. They can be expensive though.	
	What makes the difference	
	• Level 1 lists technologies with no detail of how they work or what data they give; Level 3 links each to the information it provides (territory size, presence, activity).	
	• Level 3's evaluation considers several limitations (calibration, ground-truthing, ethics, species coverage) and reaches a balanced judgement; Level 1 has one word - 'expensive'.	
	• Structure: Level 3 groups by imagery, genetics, sound, position and remote sensing; the list in Level 1 has no organisation.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Ecological features:** cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web – specific abiotic requirements (temperature, light, salinity, low turbidity) • **Importance:** quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • **Vulnerability:** narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching – slow growth so slow recovery – physical damage – overfishing and food-web effects (crown of thorns) – introduced species – dependence on mangroves and seagrass • **Conservation:** MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

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